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More constexpr containers

R0: Original proposal, presented in Albuquerque 2017.

EWG approved the direction: SF: 11 | F: 12 | N: 2 | A: 0 | SA: 0

R1: Provided initial wording.

EWG approved the proposal as presented: SF: 23 | F: 11 | N: 0 | A: 0 | SA: 0

R2: Implemented core review comments.

R3: Extends proposal for non-transient allocations.

R4: Tighten wording for "construct" and "destruct".

R5: (submitted for Kona 2019):

Introduce `mark_immutable_if_constexpr` to address some recently found issue.

Remove prohibition on virtual destructors (i.e., merge in P1077) and pseudo-destructor calls (an earlier oversight).

Replace `construct/destruct` interception by `construct_at/destroy_at` (and friends_).

Integrate San Diego CWG review notes.

Also rebase on N4778.

R6: (submitted for Cologne 2019):

Remove non-transient allocation handling because EWG did not like `mark_immutable_if_constexpr` and the problem of the R4 status quo is potentially difficult to evolve.

Add `constexpr` to the definition of `voidify` and amend the call to it in `[specialized.construct]`.

R7:

Move constraint from `construct_at` declaration into description.

Fix missing template parameters in various places.

Implemented core review comments.

Implemented library review comments.

Introduction and motivation

Variable size container types, like `std::vector` or `std::unordered_map`, are generally useful for runtime programming, and therefore also potentially useful in `constexpr` computations. This has been made clear by some recent experiments such as the [Constexpr ALL the things!](#) presentation (and its companion paper [P0810R0](#) published in the pre-Albuquerque mailing) by Ben Deane and Jason Turner, in which they build a compile-time JSON parser and JSON value representation using `constexpr`. Amongst other things, the lack of variable size containers forces them to use primitive fixed-size data structures in the implementation, and to parse the input JSON string twice; once to determine the size of the data structures, and once to parse the JSON into those structures.

We also expect variable size containers to be a necessity in the reflection and metaprogramming APIs that will emerge from the work in SG-7, which decided that the preferred direction for a standard solution would involve `constexpr`-like computation. For example, querying the template arguments of a class type might look something like:

```
std::vector<std::meta_info> args = std::meta::get_template_args(reflexpr(T));
```

Non-transient allocation

Earlier versions of this paper ([P0784r1](#) and [P0784r2](#)) proposed the changes needed for constexpr destructors and for so-called "transient constexpr allocations". Transient constexpr allocations are dynamic memory allocations occurring during a constexpr evaluation that are deallocated before that evaluation completes.

What about dynamically allocated constexpr storage that hasn't been deallocated by the time evaluation completes? There are really compelling use cases where this might be desirable. E.g., this could be the basis for a more flexible kind of "string literal" class. We therefore proposed that a non-transient constexpr allocation be a valid result for a constexpr variable initializer if:

- the result of evaluating the initializer is an object with a nontrivial constexpr destructor, and
- evaluating that destructor would be a valid core constant expression and would deallocate all the *non-transient* allocations produced by the evaluation of *expr*.

Furthermore, we specify that an attempt to deallocate a non-transiently allocated object by any other means results in undefined behavior. (Note that this is unlikely because the object pointing to the allocated storage is immutable.)

A question that arises in this context is whether the non-transient allocation is mutable between the completion of the initialization and the evaluation of the destructor. If the allocation is mutable, reading from that allocation during the destructor evaluation is meaningless and should thus not be accepted as part of a core constant expression evaluation. However, there are cases where having a mutable allocation is desirable. To permit both cases, we proposed a library function "std::mark_immutable_if_constexpr" to designate that a constexpr allocation is immutable in the context.

For example:

```
#include <memory>
#include <new>
using namespace std;
template<typename T> struct S: allocator<T> {
    T *ps;
    int sz;
    template<int N> constexpr S(T (&p)[N])
        : sz{N}
        , ps{this->allocate(N)} {
        for (int k = 0; k<N; ++k) {
            new(this->ps+k) T{p[k]};
        }
        std::mark_immutable_if_constexpr(this->ps);
    }
    constexpr ~S() {
        for (int k = 0; k<this->sz; ++k) {
            (this->ps+k)->T::~~T();
        }
        this->deallocate(this->ps, this->sz);
    }
};

constexpr S<char> str("Hello!");
// str ends up pointing to a static array
// containing the string "Hello!".
```

The constructor constexpr evaluation in this example is successful, producing an S object that points to a non-transient constexpr allocation. The constexpr evaluation of the destructor would also be successful and would deallocate the non-transient allocation. The non-transient allocation is therefore promoted to static storage.

In Kona 2019, this approach was considered too brittle, and as a result non-transient allocation was removed from the feature set.

Proposal P0784R5 as presented, for C++20.

SF: 3 | F: 4 | N: 8 | A: 5 | SA: 8

Adopt P0784R5 without non-transient allocation into C++20.

SF: 10 | F: 13 | N: 2 | A: 0 | SA: 0

Virtual constexpr destructors

In Rapperswil, Peter Dimov proposed (P1077) to let literal types have virtual destructors. That proposal passed easily:

Allowing literal types to have virtual destructors:

SF: 11 | F: 12 | N: 2 | A: 0 | SA: 0

The notion of *quasi-trivial destructor* in P1077, however, is subsumed by this paper's notion of *constexpr destructor*. It was therefore decided to merge P1077 into this paper (with P1064, which permits virtual constexpr functions already approved by WG21 and edited into the current draft working paper).

Implementation experience

The proposal in this paper (which now excludes non-transient allocations) has been implemented in the EDG compiler. However, no public deployment has been made of that implementation. Based on preliminary discussion with implementers working on Clang and MSVC, no blockers that would make this feature unimplementable or prohibitively expensive to implement have been identified.

Acknowledgments

Special thanks to Billy O'Neal for helping formulate the library wording changes.

Wording changes

Note 1: These are cumulative changes: They include the changes EWG approved for P0784r1, and build on top of that.

Note 2: The following changes enable "constexpr destructors". See further down for allocation-related changes.

Change in [basic.types] bullet 10.5.1:

— it has a ~~trivial~~constexpr destructor ([dcl.constexpr]),

Change in [expr.const] paragraph 4:

An expression *e* is a *core constant expression* unless the evaluation of *e*, following the rules of the abstract machine (6.8.1), would evaluate one of the following expressions:

— this (7.5.2), except in a constexpr function ~~or a constexpr constructor~~ ([dcl.constexpr]) that is being evaluated as part of *e*;

— an invocation of a ~~non-constexpr~~ function ~~other than a constexpr constructor for a literal class, a constexpr function, or an implicit invocation of a trivial destructor (10.3.7)~~ [Note: Overload resolution (11.3) is applied as usual — end note];

- an invocation of an undefined constexpr function ~~or an undefined constexpr constructor~~;
- an invocation of an instantiated constexpr function ~~or a constexpr constructor~~ that fails to satisfy the requirements for a constexpr function ~~or a constexpr constructor~~ (10.1.5);
- an *id-expression* that refers to a variable or data member of reference type unless the reference has a preceding initialization and either
 - it is initialized with a constant expression or
 - its lifetime began within the evaluation of *e*;
- modification of an object (7.6.18, 7.6.1.6, 7.6.2.2) unless it is applied to a non-volatile lvalue of literal type that refers to non-volatile object whose lifetime began within the evaluation of *e*

Add new paragraph after [expr.const] paragraph 4:

An object *a* is said to have *constant destruction* if:

- it is not of class type nor (possibly multi-dimensional) array thereof, or
- it is of class type or (possibly multi-dimensional) array thereof, that class type has a constexpr destructor, and for a hypothetical expression *e* whose only effect is to destroy *a*, *e* would be a core constant expression if the lifetime of *a* and its non-mutable subobjects (but not its mutable subobjects) were considered to start within *e*.

Change in [dcl.constexpr] paragraph 2:

A constexpr or consteval specifier used in the declaration of a function ~~that is not a constructor~~ declares that function to be a *constexpr function*. ~~Similarly, a constexpr or consteval specifier used in a constructor declaration declares that constructor to be a constexpr constructor.~~

Change in [dcl.constexpr] paragraph 3 bullet 1:

its return type (if any) shall be a literal type;

Add a bullet to [dcl.constexpr] paragraph 3:

if the function is a constructor or destructor, its class shall not have any virtual base classes;

Change in [dcl.constexpr] paragraph 4:

The definition of a constexpr constructor ~~whose function-body is not = delete~~ shall additionally satisfy the following requirements:

- ~~the class shall not have any virtual base classes;~~
- ~~each of the parameter types shall be a literal type;~~

~~In addition, either its function-body shall be = delete, or it shall satisfy the following requirements:~~

- ~~either its function-body shall be = default, or the compound-statement of its function-body shall satisfy the requirements for a function-body of a constexpr function;~~
- every non-variant non-static data member and base class subobject shall be initialized (10.9.2);
- if the class is a union having variant members (10.4), exactly one of them shall be initialized;

- if the class is a union-like class, but is not a union, for each of its anonymous union members having variant members, exactly one of them shall be initialized;
- for a non-delegating constructor, every constructor selected to initialize non-static data members and base class subobjects shall be a constexpr constructor;
- for a delegating constructor, the target constructor shall be a constexpr constructor.

[Example: ...]

Insert new paragraph after [dcl.constexpr] paragraph 4:

The definition of a constexpr destructor whose *function-body* is not = delete shall additionally satisfy the following requirement:

— for every subobject of class type or (possibly multi-dimensional) array thereof, that class type shall have a constexpr destructor.

Change in [dcl.constexpr] paragraph 6:

If the instantiated template specialization of a constexpr function template or member function of a class template would fail to satisfy the requirements for a constexpr function ~~or a constexpr constructor~~, that specialization is still a constexpr function ~~or a constexpr constructor~~, even though a call to such a function cannot appear in a constant expression. If no specialization of the template would satisfy the requirements for a constexpr function ~~or a constexpr constructor~~, when considered as a non-template function ~~or constructor~~, the template is ill-formed, no diagnostic required.

Change in [dcl.constexpr] paragraph 8:

The constexpr specifier has no effect on the type of a constexpr function ~~or a constexpr constructor~~.

Change in [dcl.constexpr] paragraph 9:

In any constexpr variable declaration, the full-expression of the initialization shall be a constant expression (7.7). A constexpr variable shall have constant destruction.

Change in [class.dtor] paragraph 1:

Each decl-specifier of the decl-specifier-seq of a destructor declaration (if any) shall be friend, inline, ~~or~~ virtual, constexpr, or consteval.

Add after [class.dtor] paragraph 9:

The defaulted destructor is a constexpr destructor if it satisfies the requirements for a constexpr destructor ([dcl.constexpr]). [Note: In particular, a trivial destructor is a constexpr destructor. — end note]

Note 3: The following changes enable some "constexpr new-expressions".

Modify [expr.new] paragraph 12 (the deleted part will be re-inserted below)

An implementation is allowed to omit a call to a replaceable global allocation function (16.6.2.1, 16.6.2.2). When it does so, the storage is instead provided by the implementation or provided by extending the allocation of another *new-expression*.

~~The implementation may extend the allocation of a new-expression e1 to provide storage for a new-expression e2 if the following would be true were the allocation not extended:~~

- the evaluation of *e1* is sequenced before the evaluation of *e2*, and
- *e2* is evaluated whenever *e1* obtains storage, and
- both *e1* and *e2* invoke the same replaceable global allocation function, and
- if the allocation function invoked by *e1* and *e2* is throwing, any exceptions thrown in the evaluation of either *e1* or *e2* would be first caught in the same handler, and
- the pointer values produced by *e1* and *e2* are operands to evaluated *delete-expressions*, and
- the evaluation of *e2* is sequenced before the evaluation of the *delete-expression* whose operand is the pointer value produced by *e1*.

[*Example:*

...

— *end example*]

Add new paragraph after [expr.new] paragraph 12

During an evaluation of a constant expression, a call to an allocation function is always omitted. [Note: Only *new-expressions* that would otherwise result in a call to a replaceable global allocation function can be evaluated in constant expressions (see [expr.const]). — *end note*]

Add new paragraph after [expr.new] the previously-insert paragraph (this was originally part of paragraph 12)

The implementation may extend the allocation of a *new-expression* *e1* to provide storage for a *new-expression*. *e2* if the following would be true were the allocation not extended:

- the evaluation of *e1* is sequenced before the evaluation of *e2*, and
- *e2* is evaluated whenever *e1* obtains storage, and
- both *e1* and *e2* invoke the same replaceable global allocation function, and
- if the allocation function invoked by *e1* and *e2* is throwing, any exceptions thrown in the evaluation of either *e1* or *e2* would be first caught in the same handler, and
- the pointer values produced by *e1* and *e2* are operands to evaluated *delete-expressions*, and
- the evaluation of *e2* is sequenced before the evaluation of the *delete-expression* whose operand is the pointer value produced by *e1*.

[*Example:*

...

— *end example*]

Change in [expr.const] paragraph 4:

— a pseudo-destructor call (7.6.1.10);

...

— a *new-expression* (7.6.2.4);

— a *new-expression* (7.6.2.4), unless the selected allocation function is a replaceable global allocation function (16.6.2.1, 16.6.2.2) and the allocated storage is deallocated within the evaluation of *e*;

— a *delete-expression* (7.6.2.5) unless it deallocates a region of storage allocated within the evaluation of *e*;

— a call to an instance of `std::allocator<T>::allocate(_allocator.members_)`, unless the allocated storage is deallocated within the evaluation of *e*;

— a call to an instance of `std::allocator<T>::deallocate(_allocator.members_)`, unless it deallocates a region of storage allocated within the evaluation of *e*;

Note 4: The following changes enable the use of the default allocator in constant expressions.

Add a new paragraph after [expr.const] paragraph 4:

For the purposes of determining whether an expression is a core constant expression, the evaluation of a call to a member function of `std::allocator<T>` as defined in `_allocator.members_`, where *T* is a literal type, does not disqualify the expression from being a core constant expression, even if the actual evaluation of such a call would otherwise fail the requirements for a core constant expression. Similarly, the evaluation of a call to `std::destroy_at`, `std::ranges::destroy_at`, `std::construct_at`, or `std::ranges::construct_at` is a valid core constant expression unless

- for a call to `std::construct_at` or `std::ranges::construct_at`, the first argument, of type *T**, does not point to storage allocated with `std::allocator<T>` or the evaluation of the underlying constructor call is not a core constant expression, or
- for a call to `std::destroy_at` or `std::ranges::destroy_at`, the first argument, of type *T**, does not point to storage allocated with `std::allocator<T>` or the evaluation of the underlying destructor call is not a core constant expression.

Modify [specialized.algorithms] paragraph 6 to add the `constexpr` specifier to the declaration of `voidify`:

```
template<class T>
    constexpr void* voidify(T& ptr) noexcept {
        return const_cast<void*>(static_cast<const volatile void*>
(addressof(ptr)));
    }
```

In [memory.syn] paragraph 1 and [specialized.destroy] paragraph 1 add the `constexpr` specifier to the all the declarations of `destroy_at`, `destroy`, and `destroy_n` (both in namespace `std` and in namespace `std::ranges`).

```
template<class T>
    constexpr void destroy_at(T* location);
template<class ForwardIterator>
    constexpr void destroy(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
    constexpr void destroy(ExecutionPolicy&& exec, // see 25.3.5
        ForwardIterator first, ForwardIterator last);
template<class T, class Size>
    constexpr ForwardIterator destroy_n(ForwardIterator first, Size n);
template<class ExecutionPolicy, class ForwardIterator, class Size>
    constexpr ForwardIterator destroy_n(ExecutionPolicy&& exec,
        ForwardIterator first, Size n); // see 25.3.5
```



```

        namespace ranges {
            template<Destructible T>
                constexpr void destroy_at(T* location) noexcept;
            template<no-throw-input-iterator I, no-throw-sentinel<I> S> requires
Destructible<iter_value_t<I>>
                constexpr I destroy(I first, S last) noexcept;
            template<no-throw-input-range R> requires
Destructible<iter_value_t<iterator_t<R>>>
                constexpr safe_iterator_t<R> destroy(R&& r) noexcept;
            template<no-throw-input-iterator I> requires
Destructible<iter_value_t<I>>
                constexpr I destroy_n(I first, iter_difference_t<I> n)
noexcept;
        }

```

In [memory.syn] paragraph 1, after the declarations of destroy add declarations for construct_at as follows

```

template<class T, class... Args>
    constexpr T* construct_at(T* location, Args&&... args);

namespace ranges {
    template<class T, class... Args>
        constexpr T* construct_at(T* location, Args&&... args);
}

```

and following [specialized.destroy] add a new subsection [specialized.construct] as follows:

```

template<class T, class... Args>
    constexpr T* construct_at(T* location, Args&&... args);

namespace ranges {
    template<class T, class... Args>
        constexpr T* construct_at(T* location, Args&&... args);
}

```

Constraints: The expression `::new (declval<void*>()) T(declval<Args>()...)` is well-formed when treated as an unevaluated operand.

Effects: Equivalent to:

```

        return ::new (voidify(*location)) T(std::forward<Args>
(args)...);

```

Modify [allocator.traits.members] paragraph 5 (about the construct member) as follows:

Effects: Calls `a.construct(p, std::forward<Args>(args)...) if that call is well-formed; otherwise, invokes ::new (static_cast<void*>(p)) T(std::forward<Args>(args)...) construct_at(p, std::forward<Args>(args)...) .`

Modify [allocator.traits.members] paragraph 6 (about the destroy member) as follows:

Effects: Calls `a.destroy(p)` if that call is well-formed; otherwise, invokes `p->~T() destroy_at(p).`

In [allocator.traits]/1 add constexpr to all the member functions.

```

[[nodiscard]] static constexpr pointer allocate(Alloc& a, size_type n);
[[nodiscard]] static constexpr pointer allocate(Alloc& a, size_type n,
const_void_pointer hint);
static constexpr void deallocate(Alloc& a, pointer p, size_type n);
template<class T, class... Args>
    static constexpr void construct(Alloc& a, T* p, Args&&... args);
template<class T>
    static constexpr void destroy(Alloc& a, T* p);
static constexpr size_type max_size(const Alloc& a) noexcept;

```



```
static constexpr Alloc select_on_container_copy_construction(const Alloc&
rhs);
```

In [allocator.traits.members] add constexpr to all the member function declarations.

```
[[nodiscard]] static constexpr pointer allocate(Alloc& a, size_type n);
[[nodiscard]] static constexpr pointer allocate(Alloc& a, size_type n,
const_void_pointer hint);
static constexpr void deallocate(Alloc& a, pointer p, size_type n);
template<class T, class... Args>
    static constexpr void construct(Alloc& a, T* p, Args&&... args);
template<class T>
    static constexpr void destroy(Alloc& a, T* p);
static constexpr size_type max_size(const Alloc& a) noexcept;
static constexpr Alloc select_on_container_copy_construction(const Alloc& rhs);
```

In [default.allocator]/1 add constexpr to the destructor, the copy assignment operator, and the allocate and deallocate members.

```
namespace std {
    template<class T> class allocator {
    public:
        using value_type = T;
        using size_type = size_t;
        using difference_type = ptr_diff_t;
        using propagate_on_container_move_assignment = true_type;
        using is_always_equal = true_type;
        constexpr allocator() noexcept;
        constexpr allocator(const allocator&) noexcept;
        template<class U> constexpr allocator(const allocator<U>&)
noexcept;

        constexpr ~allocator();
        constexpr allocator& operator=(const allocator&) = default;
        [[nodiscard]] constexpr T* allocate(size_t n);
        constexpr void deallocate(T* p, size_t n);
    };
}
```

In [allocator.members] paragraph 1, add constexpr to the declaration of allocate.

```
[[nodiscard]] constexpr T* allocate(size_t n);
```

In [allocator.members] paragraph 4, add constexpr to the declaration of deallocate.

```
constexpr void deallocate(T* p, size_t n);
```

In [allocator.globals] declare both comparison operators to be constexpr.

```
template<class T, class U>
    constexpr bool operator==(const allocator<T>&, const allocator<U>&)
noexcept;
```

1 Returns: true.

```
template<class T, class U>
    constexpr bool operator!=(const allocator<T>&, const allocator<U>&)
noexcept;
```

2 Returns: false.

Add to Table 17: Feature-test macros with the appropriate value:

`__cpp_constexpr_dynamic_alloc`

Add to Table 36: Standard library feature-test macros with the appropriate value:

`__cpp_lib_constexpr_dynamic_alloc` `<memory>`